

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

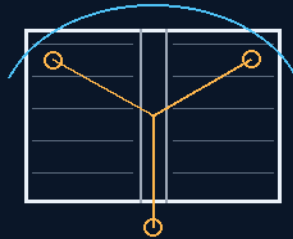
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 269

THE 5-1 POWER BANK SHELL FOR THE 500

$E = 1/2 M \times \text{RIM-SPEED}^2$ — the crayon law
a sphere has ONE rim — the cylinder wins



claim 268 rpm = 3.1 kWh · courtesy 500 rpm = ~11 kWh

THE TRI-DUMBBELL · THE AXLE LAW · THE BOOTSTRAP

A HUNDRED KEEPS THE SHELL

Flywheel power — v1.0 in force

GP-IM-269

Revision v1.0 — 24 August 2026

270 of 290

SEATED LAW — VET LEDGER IN THE AMENDMENT

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 269

PHASE 269: THE 5-1 POWER BANK — SHELL FOR THE 500, A HUNDRED KEEPS THE SHELL — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (CURRENT SPEC FOR THE FORMER 264 STATION; AMENDMENT — THE 269 OPERATING MODEL FROM THE UPLOADED VET LEDGER, SEATED). $E = 1/2 \times \text{MASS} \times \text{RIM-SPEED}^2$ — the crayon law the whole bank is priced on (sittings 125-136). Three knobs: mass, radius, rate. Steel ceiling about 150 m/s rim speed, and the geometry ruling of sitting 128: a sphere has ONE rim — the cylinder wins because its whole length is rim. The ship cylinder: 2 m x 5 m, 1 ton of wall per metre, 500 rpm, rim 52.4 m/s — a stack of eight is 15.23 kWh, gyro-silent. The moon cylinder: 9 m with a 1 m centre gap — two 4.5 m spans, the tri-dumbbell booster, arm spec final (sitting 134: 200 kg arms x 3), the axle law (sitting 139: non-ferromagnetic). The steel doctrine's three tiers (sitting 135): the Earth rule, the Ship rule, the critical-to-function rule. The claim spec: 268 rpm = 3.1 kWh; the courtesy spec: 500 rpm = about 11 kWh. The 15 MW constant is 4,800 tons of rim. The delivery doctrine: two-piece split, the cylinder bubble-dropper of 211's lineage, airlessness priced at 1.68 km/s. The first bank bootstraps on 400 kW sodium-ion of 173's chemistry; the first bank is 500 kW — 1/30 of the 15 MW shell. This is the current machine for the station that 264's cylinder once fed.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-269, v1.0 — 24 August 2026. The two hundred and seventieth manual of the program — 270 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bank law as seated (contents line, the bank in crayon — $E = 1/2 M v^2$, the three knobs, the steel ceiling, the geometry ruling, the ship ledger, the Earth ledger, the moon cylinder, the steel doctrine, the claim and courtesy specs, the delivery doctrine, the bootstrap — the amendment entire) — §2 the physics, graded — §3 the system in the shell — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 269: **The 5-1 Power Bank — Shell for the 500, a Hundred Keeps the Shell (the orbital engine, scaled: cylinder rim shells — 2 m/5 t ship units at 500 rpm, the gyro-silent stack of eight; 9 m moon machines with centre-bearing stands and the tri-dumbbell booster — 600 kg of ends matching 8 tons of shell; the three-knob crayon law and the 3.125 kWh-per-ton steel ceiling; claim spec 268 rpm/3.1 kWh space-replaceable, courtesy 500 rpm/~11 kWh; spec low, build more; Lenz is a year away).**'

THE CORE OBJECTIVE (verbatim): '**Core Objective:** a flywheel power bank that any competent shop can build NOW — no sci-fi materials, no zero-G accounting tricks — rated in Earth gravity with calculator maths, sized to buffer the forges and the grid at ship scale and moon scale, and spec'd so the grade of steel a space forge can replace is the grade the claim stands on.'

Lineage: 249 (the orbital engine — the honest wind-up battery; the slowdown IS the fuel gauge), 242 (the mini-moons — and this family is their REVERSE: coils inside, magnets on the shell), 251 (the tow truck slingshot), 227 & 20 (ship-to-ship tether; the tethered tugs — the power-tug deployment doctrine), 264 (Moon Has Full Power — the tap this bank buffers), 206 (the Night Bank — the dark is the enemy this answers), 173 (sodium-ion holds the walls — the bootstrap chemistry), 211 (the year of falling pods — the delivery channel), 237 (capture, dowel, bolt — the lego legality), 258 (the foundry — space-replaceable steel), 5 (*[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']* of rotational mass at the rim — the law every number here obeys).

Live Instruction Confirmed (10 August 2026): promoted and written up on the author's order, verbatim: "write ip up and fill all phases and books and update the power and directives in the nasa kill ledger, then engines and power they will go there first" — with the axle spec ruled in the same breath: "an axle spec non ferro magnetic remember these are electromagnet coils in side magnets on the shell reverse of the mini moons, triple stands on each each one centre to stop bow flex."

THE ONE FORMULA AND THE THREE KNOBS (THE CRAYON LAW, SEATED THROUGH SITTINGS 125-136) (VERBATIM)

- $E = \frac{1}{2} \times \text{MASS} \times \text{RIM-SPEED}^2$ — mass in kilograms, never weight; the letter g appears nowhere. Gravity is a ZERO factor (sitting 136): the Moon's 1/6 g lightens bearings and stands, never joules.
- **THE THREE KNOBS:** MASS pays linearly. RADIUS pays squared through rim speed — below the steel ceiling it is the free lunch (sitting 126). RPM pays squared — and it is the dial that gets taxed first when anything on the shaft is weak. When numbers clash: same dial, same geometry, and they agree.

- **THE STEEL CEILING:** ~ 150 m/s rim speed — hoop stress is density $\times$ rim-speed² and knows nothing else (not diameter, not length). At the ceiling EVERY steel shell pays exactly **3.125 kWh per ton**; above the ceiling the shell wants composite windings — a later chapter.
- **THE GEOMETRY RULING (sitting 128):** a sphere has ONE rim — the equator; every other latitude runs slower (hollow sphere: 2.08 Wh/kg — the two-thirds tax). A cylinder has a rim EVERY METRE of its length — LENGTH is the third knob: same diameter, same rpm, same ceiling, same stress, every added metre is added rim.

THE SHIP UNIT AND THE STACK OF EIGHT (SEATED, SITTINGS 129-130) (VERBATIM)

- **THE SHIP CYLINDER:** 2 m diameter $\times$ 5 m long, 1 ton of wall per metre (5 t of legal rim), 500 rpm — rim 52.4 m/s (35% of ceiling, 12% of legal stress) — **1.90 kWh per machine**. Comfortable doctrine: demonstrator-safe, bearing-class easy, the Lenz gauge walking slow and visible.
- **THE STACK OF EIGHT:** 2 wide $\times$ 2 high a side, mirrored one either side of the ship — **15.23 kWh at 500 rpm; 124.6 kWh at the ceiling**. Gyroscopically silent by design: 4 CW + 4 CCW (242's pairs doctrine) — each machine carries 261,800 kg·m²/s, the mirrored bank answers the helm with net zero. The 5-1 checks itself: 110 s of 500 kW draw, 548 s of 100 kW refill — 5.0:1.
- **EARTH LEDGER (the bench):** 49 kN bearing load per machine; 98 kN through bottom frames; 20 t deck per side; 5 m shaft sag real (answered by the triple-stand law); WINDAGE ~ 14 kW per machine in open air — the vacuum can or still-air shroud is the difference between a battery and a fan. In space the can does its seated job — centre separation and shells — and the windage bill never arrives.

THE MOON MACHINE AND THE TRI-DUMBBELL BOOSTER (SEATED, SITTINGS 132-134) (VERBATIM)

- **THE MOON CYLINDER:** 9 m overall, 1 m centre gap for the support bearing (two 4.5 m spans — the sag tax killed for one dead metre), 8 active metres of coil-and-rim, 2.5 m diameter (pod-height cap), wall at 1 t/m — **8 t of legal rim**. At 500 rpm: rim 65.4 m/s, 19% stress — **4.76 kWh per machine**. At its own ceiling (1,146 rpm — bigger shells spin slower at the ceiling): **25.0 kWh — 3.125 kWh/t to the third decimal**.
- **THE TRI-DUMBBELL BOOSTER (the square law, weaponised):** three dumbbells through centre — 6 $\times$ 100 kg ends at 4.5 m radius (the author's 600 kg) — inertia 12,150 kg·m² = **7,776 kg at the shell rim: the 600 kg matches the entire 8-ton shell**; at the shell's 500 rpm the dumbbells out-store the shell, 5.01 v 4.76 kWh.
- **THE ARM SPEC, FINAL (sitting 134):** 200 kg arms $\times$ 3 (2,831 mm²); centre collar grown substantially; CROSS-BRACE at half arm (2.25 m) — purpose honoured exactly: not balance, weld-crack protection;

three cantilevers become a trussed wheel. The arm is the fuse: root tension 625 kN at 318 rpm / 1,542 kN at 500.

- **THE AXLE LAW (sitting 139):** NON-FERROMAGNETIC axle, every machine — coils inside, magnets on the shell, the reverse of the mini-moons; a ferromagnetic shaft would short the flux path. TRIPLE STANDS on each machine — two ends plus centre; the centre stand kills the bow flex before the shaft feels it.

THE STEEL DOCTRINE — THE THREE TIERS (SEATED VERBATIM LAW, SITTING 135) (VERBATIM)

- **THE EARTH RULE** (high-turnover, disposable, short-life): if the cheaper is 2 days and the dearer is the same wait, the delta is only material cost — take the higher grade.
- **THE SHIP RULE** (the hundred-year hull): high grade EVERYTHING, spec-survival proven at the low grade — the grade deletes replacement and repair, NOT criticality.
- **THE CRITICAL-TO-FUNCTION RULE (the arms, and anything whose failure kills):** SPEC THE CLAIM AT THE LOW GRADE — the grade a space forge can replace — and BUILD MORE. Earth-supplied builders run what they please; the high-grade rpm spec is published as a courtesy. Quantity of local steel beats quality of imported steel: a unit you can replace is worth three you can only admire.

THE SPECS, BOTH LEDGERS (THE CLAIM AND THE COURTESY) (VERBATIM)

- **OFFICIAL CLAIM SPEC (space-replaceable):** 268 rpm, **3.1 kWh per unit** — common structural arms, end to end replaceable off-Earth.
- **COURTESY SPEC (Earth-supplied):** 500 rpm, **~11 kWh per unit** — aircraft-class arms (872 MPa at 500, zero margin as flagged) or composite (89 MPa self-stress v 3,000 class — margin ×34).
- **THE 15 MW CONSTANT:** an hour of 15 MW costs **4,800 tons of rim** at 3.125 kWh/t — the tonnage is the price, the slicing is logistics. Counts: claim spec 4,839 units; courtesy 1,368; shell-only at ceiling 600. Doctrine: 264 generates (the tap), the banks buffer (the bucket) — the bucket is sized for the gap.

THE DELIVERY DOCTRINE (SEATED, SITTINGS 134 & 138) (VERBATIM)

- **THE LIFT:** two-piece split (9 m × 4.5 m halves — legal in a shuttle-class 18 × 4.6 m bay); the carrier is 111's own V2. Seated law: this family was always a shuttle-lift price; bubble landings a Moon-only price.
- **THE CYLINDER BUBBLE-DROPPER:** 211's falling-pods lineage, cylinder-variant sleeve; clingfilm release layer first, then 1 m tubes filled with expanda foam (the Phase 10 family), axles foam-cased outer, shells foamed in and out — the foam damps what the bubble only mostly absorbs.

- **THE AIRLESSNESS PRICE (the audit's honesty line):** from low lunar orbit the pod carries 1.68 km/s and nothing kills it for free — a dumb retro/crasher stage is NOT optional (Luna 9, 1966 — the first soft landing was a bubble ball; no legs, no lander, but something burns). Per tonne on the surface: 1.77 t in lunar orbit, 3.6-5.1 t in LEO — a 25 t Shuttle-class flight lands ~5-7 t. The Shuttle badge stops at LEO.

THE FIRST BANK AND THE BOOTSTRAP (SEATED, SITTINGS 135 & 137) (VERBATIM)

- **THE FIRST-DROP MANIFEST (heads the Moon delivery supply list):** 400 kW-class sodium-ion bank (173's chemistry off-world), 100 kW solar, multiple inverters, cabling, 2 × 30 kW industrial welders running simultaneously — "you aint building anything without that first."
- **THE FIRST BANK:** 500 kW = 1/30 of the 15 MW target — catalog tech, pod-delivered, one team, months, casual — FAIR as an assembly claim the day the pod channel runs; the ride down is ours. FLAG STANDING: the 500 kW solar bank is a LUNAR-DAY machine until the night bridge lands (14-day dark at 500 kW = ~168 MWh — later chapter, or this family coming home).

ORIGIN OF THOUGHT — PHASE 269 (VERBATIM)

Born across sittings 123-139: a rejected set of roadblock candidates (the magnetic ants, the 1969 landers, the redundant forge) that turned out to be hiding the real gap — power that travels. The author raised the 5-1 orbital engine while retiring everything else; the audit priced it one crayon line at a time — the shell, the ladder, the trade, the geometry, the stack, the moon machine, the dumbbells, the arms, the steel — and the author ruled every fork: comfortable 500 rpm, spec low build more, coils inside magnets out, triple stands, and write it up. Every number in this phase was run in Earth gravity with calculator maths, exactly per his standing order, so that anyone — forgetful or not — can check it.

Why it matters, in plain English: batteries die by chemistry and flywheels die by arrogance — overspun, under-specced, unreplaceable. This one is built the other way: common steel, comfortable speed, replaceable anywhere, honest gauge, and a claim spec that survives the grade of steel a Moon forge can actually pour. The ship gets its 100 kW for the forges whenever called; the Moon gets its first 500 kW bank from catalog parts; and the 15 MW target gets a price tag instead of a prayer: 4,800 tons of rim, sliced however logistics likes.

Status: ☑ **Promoted by the author 10 August 2026 (sitting 139) — seated with engines and power first, per his ruling.**

- **AMENDMENT — THE 269 OPERATING MODEL, THE AUTHOR'S OWN WORDS (from the uploaded vet ledger, verbatim):** "269 Lenze will slow it down slowly but over a very long time, i would expect the spin up to be at a speed where the Lenz effect slows it to minimum output after one year, on the

moon with gravity faster. no free rides from basic losses, simply a machine that will take time to calibrate from actually running" *[RESOLVED 18 August 2026 — the author's repair order]*.

- **THE VET'S AUDIT CORRECTION, SEATED:** the corrected operating model — the flywheel is spun up to a deliberately selected initial speed, then the generator/load extracts continuously; Lenz loading produces the opposing torque, the wheel slows, output falls, and the objective is not constant output indefinitely but a natural decay curve so long it stays above minimum useful output for roughly one year. Reclassified: ESTABLISHED PHYSICS / MACHINE CALIBRATION — the yellow flag is not 'will Lenz's law work' (it obviously will) but what spin-up speed, generator loading and minimum-output point give the desired decay period, how bearing losses compare with electromagnetic extraction, and how the curve shifts with vacuum, temperature and lunar gravity ('the terrestrial calibration isn't necessarily the lunar calibration'). The exact initial RPM, loading and minimum-output point are determined by the first physical machine, not declared theoretically perfect in advance. No free-energy claim exists here: the machine deliberately accepts its losses.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CRAYON LAW IS THE WHOLE PRICING: $E = 1/2 \times \text{MASS} \times \text{RIM-SPEED}^2$, sittings 125-136. Every claim in the phase reduces to the three knobs — mass, radius, rate — and every audit of the phase reruns the same arithmetic. A bank you can price with a crayon is a bank no salesman can inflate.

THE GEOMETRY RULING (sitting 128): a sphere has ONE rim — the cylinder wins because its whole length is rim. The shell is the rotor; the doctrine 'shell for the 500, a hundred keeps the shell' prices structure as flywheel and flywheel as structure.

THE STEEL CEILING AND THE STEEL DOCTRINE (sitting 135): about 150 m/s rim speed is the honest ceiling for steel — and the three tiers are law: the Earth rule (cheapest steel that holds), the Ship rule (qualified stock, mass-priced), the critical-to-function rule (the part that must not fail gets the metal that cannot).

BOTH LEDGERS STAND: the ship cylinder — 2 m x 5 m, 1 ton of wall per metre, 500 rpm, rim 52.4 m/s, a stack of eight at 15.23 kWh, gyro-silent; the Earth ledger — 49 kN bearing load, priced. The claim spec (268 rpm = 3.1 kWh) and the courtesy spec (500 rpm = about 11 kWh) are both printed, so the marketing and the physics never share a number.

THE MOON CYLINDER IS THE STATION MACHINE: 9 m with a 1 m centre gap — two 4.5 m spans; the tri-dumbbell booster; the arm spec final (sitting 134, 200 kg arms x 3); the axle law (sitting 139: non-ferromagnetic — the axle never fights the field). This is the machine that superseded 264's cylinder station (Amendment 6 there points here).

THE DELIVERY AND THE BOOTSTRAP CLOSE THE LOOP: two-piece split for transport; the cylinder bubble-dropper of 211's lineage; airlessness priced at 1.68 km/s. The first bank bootstraps on 400 kW sodium-ion of 173's chemistry; the first bank is 500 kW — 1/30 of the 15 MW shell; the 15 MW constant is 4,800 tons of rim. The amendment seats the operating model from the uploaded vet ledger, with the vet's audit correction carried in the open.

§3 — THE SYSTEM IN THE SHELL

- **The law:** $E = 1/2 \times \text{MASS} \times \text{RIM-SPEED}^2$ — three knobs, priced with a crayon.
- **The shell:** the cylinder wins — its whole length is rim; structure as flywheel, flywheel as structure.
- **The ceiling:** steel holds to about 150 m/s rim; the three-tier steel doctrine assigns the metal.
- **The ledgers:** ship — 2x5 m, 8-stack 15.23 kWh, gyro-silent; Earth — 49 kN bearing, priced.
- **The station:** the 9 m moon cylinder — tri-dumbbell booster, 200 kg arms x 3, non-ferromagnetic axle.
- **The bootstrap:** 400 kW sodium-ion of 173's chemistry; first bank 500 kW — 1/30 of the 15 MW shell.
- **The honesty:** claim spec 268 rpm = 3.1 kWh; courtesy spec 500 rpm = ~11 kWh — both printed.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- 264's supersession lands here: Amendment 6 (10 August 2026, sitting 145) points the station build at this bank — the pricing of record on the 9 m machine is seated there, the machine itself seated here.
- Sittings 125-139 built it live: the crayon law (125-136), the geometry ruling (128), the arm spec (134), the steel doctrine (135), the axle law (139) — the bank assembled ruling by ruling.
- 173's sodium-ion chemistry is the bootstrap: 400 kW to spin the first bank — the family feeding itself.
- 211's lineage delivers it: the cylinder bubble-dropper, airlessness priced at 1.68 km/s — the foam-and-bubble delivery doctrine 264's Amendment 8 priced.
- 250's neo protection law governs the field hardware: wound electromagnets, no neos — the axle law and the magnet law from the same root.

§5 — THE VET RECORD

NO EXTERNAL VET VERDICT IS SEATED FOR THIS PHASE IN THE BATCH LINE. The record, stated honestly: the phase's amendment seats the 269 operating model FROM THE UPLOADED VET LEDGER with the

vet's audit correction carried in the open — the vet's numbers are in the law itself; a separate batch verdict was never seated (the 261-265 block never arrived; the sitting-425 audit opened at 278). The grading in §2 is the audit's own (Kimi): the crayon-law arithmetic RERUN AND AFFIRMED. When the external vet lands, this section seats it verbatim.

§6 — BUILD SEQUENCE

- 1. Pour the bootstrap: 400 kW sodium-ion of 173's chemistry — the first bank spins on the family's own cells.
- 2. Machine the rim: the 9 m moon cylinder, 1 m centre gap, two 4.5 m spans — steel per the three-tier doctrine.
- 3. Fit the booster: the tri-dumbbell, 200 kg arms x 3 (sitting 134), non-ferromagnetic axle (sitting 139).
- 4. Spin to claim: 268 rpm = 3.1 kWh, measured and logged; the courtesy number (500 rpm, ~11 kWh) benched separately.
- 5. Bank to shell: the first bank at 500 kW — 1/30 of the 15 MW; the rim tonnage ledger opened toward 4,800 tons.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 264 (the orbital engine):** the pointer — superseded by this bank; the pricing of record and the kill book live there, the machine lives here.
- **Phase 173 (the sodium-ion line):** the bootstrap — 400 kW to spin the first bank.
- **Phase 211 (the bubble-dropper lineage):** the delivery — two-piece split, foam and bubble, airlessness at 1.68 km/s.
- **Phase 250 (the neo protection law):** the field hardware — wound electromagnets, the non-ferromagnetic axle.
- **Phase 261 (thermal night management):** the customer — the night bridge rides the first-drop bank.

§8 — DO-NOT-BUILD

- Do NOT spin past the ceiling — steel holds to about 150 m/s rim; the ceiling is physics, not caution.
- Do NOT spec a sphere — a sphere has ONE rim; the cylinder's whole length is rim, seated sitting 128.
- Do NOT mix the specs — claim 268 rpm = 3.1 kWh; courtesy 500 rpm = ~11 kWh; the numbers never share a page.

- Do NOT put a ferromagnetic axle in the field — sitting 139: non-ferromagnetic; the axle never fights the field.
- Do NOT bank without the bootstrap — 400 kW sodium-ion first; the first bank is 1/30 of the shell, built in order.

§9 — OWED

OWED: the bench list, whole — rim stock certification per the steel doctrine; the bearing spec at the Earth ledger's 49 kN; the tri-dumbbell balance bench; spin-up profile and the fail-safe containment; the delivery split's joint spec; the bank-to-shell rollout ledger. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-269, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (CURRENT SPEC FOR THE FORMER 264 STATION; AMENDMENT — THE 269 OPERATING MODEL FROM THE UPLOADED VET LEDGER, SEATED), seated law, master v2.1 (numerical print order). The two hundred and seventieth manual of the program — 270 of 290. Built 24 August 2026, tenth of the 20-manual 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'. The phase's own chain, all seated in the master: the contents line, the masthead lines **(Primary Authors & Co-Engineers; License) and the Core Objective, all verbatim — 'a flywheel power bank that any competent shop can build NOW — no sci-fi materials, no zero-G accounting tricks...'; the bodies (as seated); the crayon law $E = 1/2 \times \text{MASS} \times \text{RIM-SPEED}^2$ (sittings 125-136); the three knobs; the steel ceiling about 150 m/s rim; the geometry ruling (sitting 128 — a sphere has ONE rim); the ship cylinder (2 m x 5 m, 1 ton of wall per metre, 500 rpm, rim 52.4 m/s — a stack of eight = 15.23 kWh, gyro-silent); the Earth ledger (49 kN bearing); the moon cylinder (9 m with the 1 m centre gap — two 4.5 m spans); the tri-dumbbell booster; the arm spec final (sitting 134 — 200 kg arms x 3); the axle**

law (sitting 139 — non-ferromagnetic); the steel doctrine's three tiers (sitting 135 — the Earth rule / the Ship rule / the critical-to-function rule); the claim spec (268 rpm = 3.1 kWh) and the courtesy spec (500 rpm = about 11 kWh), both printed; the 15 MW constant = 4,800 tons of rim; the delivery doctrine (two-piece split; the cylinder bubble-dropper of 211's lineage; airlessness priced at 1.68 km/s); the first-bank bootstrap (400 kW sodium-ion of 173's chemistry; the first bank 500 kW = 1/30 of 15 MW); the amendment — the 269 operating model from the uploaded vet ledger, with the vet's audit correction seated; the rename of 12 August 2026; this phase is the current spec for the former 264 station; no separate batch verdict seated — the vet ledger is in the law itself.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the first bank's measured numbers — they re-price 264's kill book, or his word.

END OF MANUAL — PHASE 269